

Two ways past the wall

Extending clifft beyond its 2^k limit

A · Stabilizer-rank backend

change the *representation*:

pay for magic, not for dimensions

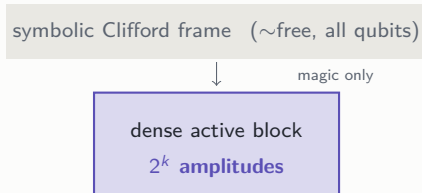
B · GPU shot batching

change the *hardware use*:

run thousands of shots abreast

clifft in thirty seconds

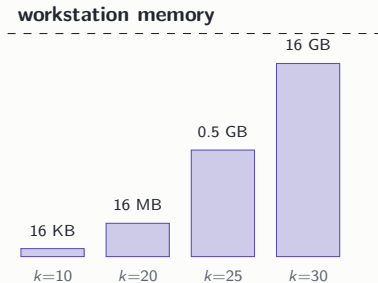
- Clifford gates are “easy”: clifft tracks them **symbolically**, almost for free, however many qubits.
- Only genuine *magic* (T gates) forces it to hold real amplitudes: a dense array of 2^k complex numbers, the **active block**.
- k counts how much magic is live at once — usually far smaller than the qubit count. That is why clifft is fast.



One engine, two tiers: symbolic where physics allows it, dense only where it must be.

The wall: 2^k doubles with every step

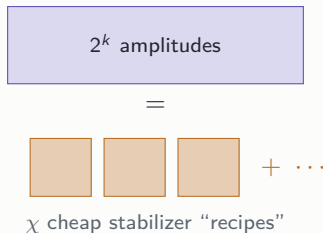
- Memory doubles per unit of k :
 $k = 30 \Rightarrow 16 \text{ GB}$, $k = 40 \Rightarrow 16 \text{ TB}$.
- clifft's compiler already fights hard for small k (measurement-driven reduction)
— circuits that still land at large k are *genuinely* dense.
- So the remaining question is not
“compile better” but: **what do we do**
when 2^k itself is the cost?



Two independent bets on what to do about it — one mathematical, one architectural.

A · The idea: pay for magic, not for dimensions

- States with *little* magic are secretly simple: they are a **short sum of “free” stabilizer states** — the states Clifford simulators handle natively.
- Store χ small recipes (a few KB each) instead of 2^k raw numbers. Like describing an image as a few overlapping patterns instead of listing every pixel.
- Clifford gates update all recipes cheaply; only T gates grow their number; **measurements shrink it back.**



The cost driver becomes the amount of magic t , not the dimension 2^k .

A · How it fits: one compiler, two executors

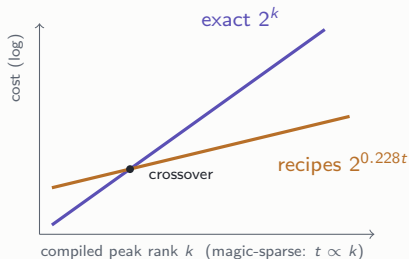


- Both cost exponents — 2^k dense, $2^{0.228t}$ backend (why 0.228: next slide) — are known **before anything runs**: dispatch is a one-line comparison, per program.
- **Wired, not just proposed**: the backend runs clifft's *optimized* circuits — compilation absorbs every Clifford and halves the live T -count (validated to 10^{-13}).
- The backend is an explicit opt-in (δ), never a silent substitute: clifft's exactness contract stays intact.

clifft compiles; the cheaper engine runs — on the same optimized circuit.

A · Why it should work

- There is an optimal way to split a T gate so the number of recipes grows only as $2^{0.228t}$ — far slower than 2^k when magic is sparse.
- Accepting a small, **tunable** error δ lets you keep only a random sample of recipes ($\sim 1/\delta^2$ of them). Measured accuracy is *predictable*: error = 0.50δ , on the nose.
- Everything stays in the language cliff already speaks: same frame idea, same measurement collapse.



Trade exactness you can measure for reach you cannot otherwise have.

A · It generalizes: any Clifford+ T circuit

- The trick for circuits with magic *everywhere*: teleport each in-line T onto its own helper qubit (the same adaptive gadget real hardware uses) — then all magic sits in **one up-front layer** again.
- So the sampler keeps its best properties on *any* circuit: **no penalty for interleaving**, and the normalization stays exactly known — the expensive estimation step never enters.
- Tested at scale on circuits whose answer is known *by construction* (hidden shift: the output is the hidden string, always): at up to **96 total qubits**, the backend returns 0.90–0.99 of the true probability 1, in seconds.
- Honest finding: clifft's compiler crushes many such circuits to tiny cores and answers in microseconds — “who wins” is a *rule*, not a race: exactly what the compile-time dispatch encodes.

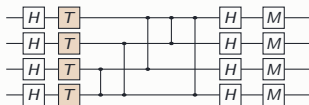
Generality at the product-magic price — validated end to end against exact references.

A · Recycled magic: pay per episode, not per program

- Long computations **recycle** their magic: the active block grows, is measured out, grows again (“episodes”). Total magic grows with runtime; **per-episode magic stays bounded**.
- Run one episode at a time, collapse *exactly* at each boundary: cost $2^{\text{per-episode magic}}$ instead of $2^{\text{total magic}}$, **zero extra error**. Measured: 72 qubits, 72 T's, ≤ 400 terms — whole-circuit would need 2^{72} .
- Compressing *below* that also works, gently: at $10\times$ fewer terms the same run returns a 10^{-22} record probability **within 41%** in a minute (unbiased; averaging sharpens it).
- Honest, measured boundary: for full records, harder compression degrades sharply with episode count — the dispatcher's **four strategies** price exact-per-episode accordingly.

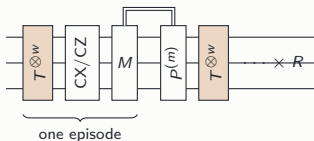
The circuit's rhythm — magic in, magic measured out — is itself a computational resource.

A · The circuits behind the numbers



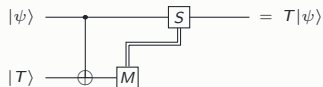
Dense IQP — the stress test

magic on every qubit; *compiles* to peak rank $n/2$ (crossover, frontier)



Magic conveyor — recycled magic

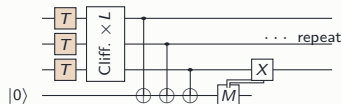
w T 's in, measured out with feedforward, R rounds (episodic)



T-gadget — the generality trick

in-line $T \rightarrow$ ancilla + measurement + conditional S : any circuit

becomes one magic layer (hidden shift, 96 qubits)



Cultivation patch — adaptive

syndrome rounds + logical Clifford traffic while magic accumulates (the

frame's home turf)

A · Evidence (measured, against cliff's real fast path)

- Every component validated to **machine precision** against exact references (incl. an independent checker we built).
- Head-to-head on circuits that compile to *genuinely large* blocks (peak rank $\approx n/2$): the backend overtakes exact cliff's at $n \approx 34$ (coarse) and $n \approx 48$ (8% error).
- Past $n \approx 62$ it is the **only method that runs at all**.
- External exact tools on the same circuits: QuiZX needs $25\times$ longer by $n=56$; Quokka# (model counting) **times out beyond** $n=24$.

$n = 72$	memory	time
any exact method	~ 1 TB	—
rank backend	1.9 GB	267 s

Accuracy dial, measured:
error = 0.50δ across the whole sweep
(halve the error $\Rightarrow 4\times$ the work).

The niche is real — and honestly bounded: below the crossover, exact cliff's stays the right tool _{10 / 16}

A · Where it wins, where it doesn't

Small peak rank	Large peak rank, sparse magic	Adaptive, Clifford-heavy
Not qubit count: clifft compiles most circuits, however wide, to a <i>small</i> block — then it is faster <i>and</i> exact. Keep it.	The niche. Even the <i>compiled</i> block breaks the memory wall, <i>T</i> -count low: the only feasible method (approximate, dial-controlled).	Real protocols win: cultivation-style patches 4–16×, teleported- <i>T</i> gadgets $\sim 2\times$; bare syndrome extraction sits at the boundary. The standard open-source tool cannot run these circuits at all.

- Also measured where it *loses*: with no logical Clifford traffic the frame's bookkeeping costs up to $\sim 2\times$ — the advantage is a measured phase boundary, not a slogan.

Status: prototype validated; honest benchmarks, natural workloads, external baselines done.

B · The idea: don't run one shot faster — run 4,096 abreast

- A cliff gate just *streams* a small array (16 KB–16 MB). One such array cannot keep a GPU busy: **a stadium choir asked to sing one sentence.**
- But cliff's workloads are Monte-Carlo: **thousands of shots of the same program**, today run one after another.
- Put $B=4,096$ shots' arrays on the GPU side by side and apply every gate to *all of them* in one launch.

CPU today: single file



GPU: all at once



one instruction, B statevectors

Batching doesn't make the GPU faster — it makes the problem big enough for the GPU to be fast.

B · Why it should work (1): shots march in lockstep

- The shape of a shot — which operation runs when, and how big the array is at every step — is **fixed at compile time and identical for every shot**. clifft's compiler already writes this schedule down.
- What differs per shot (measurement outcomes, noise) is just **a few numbers**, which enter the shared computation as per-shot parameters — like SIMD lanes, one lane per shot.
- So the classic objection to batching — “shots diverge” — *does not apply to clifft*. Its structure is unusually batchable.
- And nobody serves this niche: the two existing batched-GPU simulators both **fall back to one-shot-at-a-time** exactly when circuits measure mid-flight — clifft's home turf.

The hardest architectural question resolves in the design's favor before any GPU code is written.

B · Why it should work (2): the speed limit is bandwidth — and GPUs have $7\times$ more

- These operations do almost no arithmetic per byte: on *any* machine they run at the speed of memory.
 - Measured: clifft on 11 CPU cores is only $2.6\times$ one core — **the bus is the limit, not the cores.**
 - A GH200's memory is $\sim 7\times$ wider than a server CPU's. Once the batch saturates it: **$5\text{--}10\times$ more shots per second**, honestly bounded.
-

memory bandwidth (TB/s)



server CPU ~ 0.5



GH200 GPU ~ 3.5

big published “ $100\times$ ” numbers use weak baselines — we keep a refuted-claims list and quote the physics.

The claim is deliberately modest — which is why we can trust it.

B · The one risk, and the \$20 experiment that settles it

- Mid-circuit measurements need a decision from the CPU: results cross to the host, outcomes cross back. Each crossing costs microseconds — **the same scale as the small kernels themselves**.
- Two reasons it should survive: the crossing is paid **once per batch** (nanoseconds per shot at $B=4,096$), and the GH200's CPU-GPU link is built precisely to make it cheap.
- A microbenchmark faithful to cliff's kernels — *including complete measurements with the round-trip* — is built, validated, and waiting. GH200 time: \$2/hour, self-serve.

round-trip measured cheap **go**: batched backend; first target = replay-style queries (no round-trips at all)

round-trip eats the win **no-go**, cleanly — or on-device sampling is the follow-up

Either answer is cheap, quantitative, and final — that is what the microbenchmark is for.

How the two bets fit together

A · Stabilizer rank extends *reach*

circuits cliff cannot hold at all today —
approximate, with a measured accuracy dial

B · GPU batching extends *throughput*

the circuits cliff already runs —
many more shots per second, exact as ever

- Both work *because of* cliff's structure, not despite it: the symbolic frame, the small active block, and the compile-time shot schedule are exactly what decompositions and batching need.
- Both were reviewed adversarially; every headline number above survived a fair baseline.

Next: A — benchmarks complete; write it up. B — the GH200 run, then batched replay queries.